

# Autodesk® Scaleform®

## MMO

Scaleform 4.3 MMO  
3A

PC MMORPG

Nate Mitchell, Prasad Silva  
1.00  
2012 1 19

## **Autodesk® Scaleform® 4.3**

© 2013 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk 123D, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo), BIM 360, Built with ObjectARX (design/logo), Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, Design Server, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, Exposure, Extending the Design Team, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, Freewheel, GDX Driver, Glue, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, Map It, Build It, Use It, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, U-Vis, ViewCube, Visual, Visual LISP, Vtour, WaterNetworks, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

### **Disclaimer**

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

## Autodesk Scaleform

---

MMO

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

[www.scaleform.com](http://www.scaleform.com)

[info@scaleform.com](mailto:info@scaleform.com)

(301) 446-3200

(301) 446-3199

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| <b>1</b> | .....                                                                  | <b>1</b>  |
| 1.1      | .....                                                                  | 1         |
| <b>2</b> | .....                                                                  | <b>3</b>  |
| 2.1      | .....                                                                  | 3         |
| 2.2      | .....                                                                  | 3         |
| 2.2.1    | (Nameplate) (Inventory) (Action<br>Bar) (Window Menu) (Chat Log) ..... | 3         |
| 2.2.2    | .....                                                                  | 4         |
| 2.2.3    | .....                                                                  | 5         |
| 2.2.4    | .....                                                                  | 5         |
| 2.2.5    | .....                                                                  | 6         |
| 2.2.6    | .....                                                                  | 6         |
| 2.2.7    | .....                                                                  | 7         |
| 2.2.8    | / .....                                                                | 8         |
| 2.2.9    | .....                                                                  | 9         |
| <b>3</b> | .....                                                                  | <b>10</b> |
| 3.1      | C++ .....                                                              | 10        |
| 3.1.1    | C++ .....                                                              | 10        |
| 3.2      | Flash .....                                                            | 11        |
| 3.2.1    | MMOKit.flas .....                                                      | 12        |
| 3.2.2    | .....                                                                  | 12        |
| 3.2.3    | .....                                                                  | 13        |

1

Scaleform MMO

Autodesk®

Scaleform®

(UI)

CLIK

MMO

(Inventory)

(Paper Doll)

(Drag and Drop)

(Windowing)

(Asset)

ActionScript

C++

/



1 MMO

MMORPG

1.1

## MMO

## UI

- 1.
2. / (Spell / Ability Book)
- 3.
- 4.
- 5.
6. (Quick-Action)
7. (Cast Bar)
8. (Window)
9. (Chat Log with Tabs)

- 1.
- 2.
- 3.
- 4.

## 2

### 2.1

(Demo)

- *Apps/Kits/MMO/* - MMO C++
- *Bin/Data/AS3/Kits/MMO/* - Flash ActionScript
- *Projects/Win32 /{Msvc80, Msvc90, or Msvc10}/Kits/MMO/* – Windows  
Visual Studio 2005/2008/2010

*Bin/Kits/MMO* Windows  
MMOKit.exe *Scaleform SDK*

Windows *Projects/Win32/Msvc80/Kits* *Msvc90/Kits* *Msvc10/Kits* Scaleform  
4.2 Kits.sln “Working”  
directory” *Bin/Data/AS3/Kits/MMO*

### 2.2

**2.2.1** (Nameplate) (Inventory)  
(Action Bar) (Window Menu) (Chat Log)

## 2

(Player) Target  
Nameplate

### 2.2.2

3

(Backpack)

4

(Item)

MMO

2.2.3

5        1    2  
          12

“(Cooldown)

2.2.4



6

“ ”

2.2.5

7

MMORPG

/

MMO

/

HTML

2.2.6

8

/

MMO

2.2.7



9

(Paper Doll)

(Equipment Manager)

ScrollingList

—

ScrollingList  
TextField

HTML

“Head”

“Helm of Redemption”

“Head”

2.2.8 /

10 /

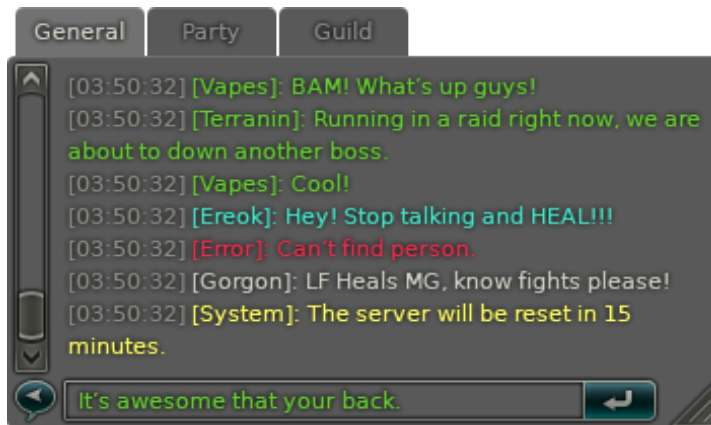
/

“ ”

“ ” “ ”

ScrollingList

## 2.2.9



11

MMO

(Chat Log)

“ ”

/s (Say) /1 (World) /p (Party) /g

(Guild)

/t <player name> <message>

“ ”

## 3

MMO

## Flash UI

## C++

# C++

MMO

—

•

### 3.1 C++

# C++

1.

2.

Flash

## MVC

Flash

## ActionScript

## Scaleform

Scaleform Direct Access API  
(Menu Kit)                      MMO

MMO

UI

HUD

## Direct Access API

## ActionScript

## UIComponent

## UIComponent

## UIComponent

### 3.1.1 C++

# C++

*Apps/Kits/MMO/*

### 3.1.1.1

- **MMOKitDemo.cpp** – Scaleform
- **Game.cpp** –

### 3.1.1.2 C++    ActionScript

- **GameUIDataBindings.cpp** – GameUIAdapter Flash  
UIComponent C++ Flash

- **GameUIAdapter.cpp** – GameUIAdapter

### 3.1.1.3

- **GameClient.cpp** – GameClient GameClient  
GameUIAdapter GameUIAdapter UI
- **GameServer.cpp** – MMORPG MMO  
GameClient UI Flash UI GameClient

### 3.1.1.4

- **GameUIRuntimeAtlasing.cpp** – GPU  
(Draw Call) MMO

## 3.2 Flash

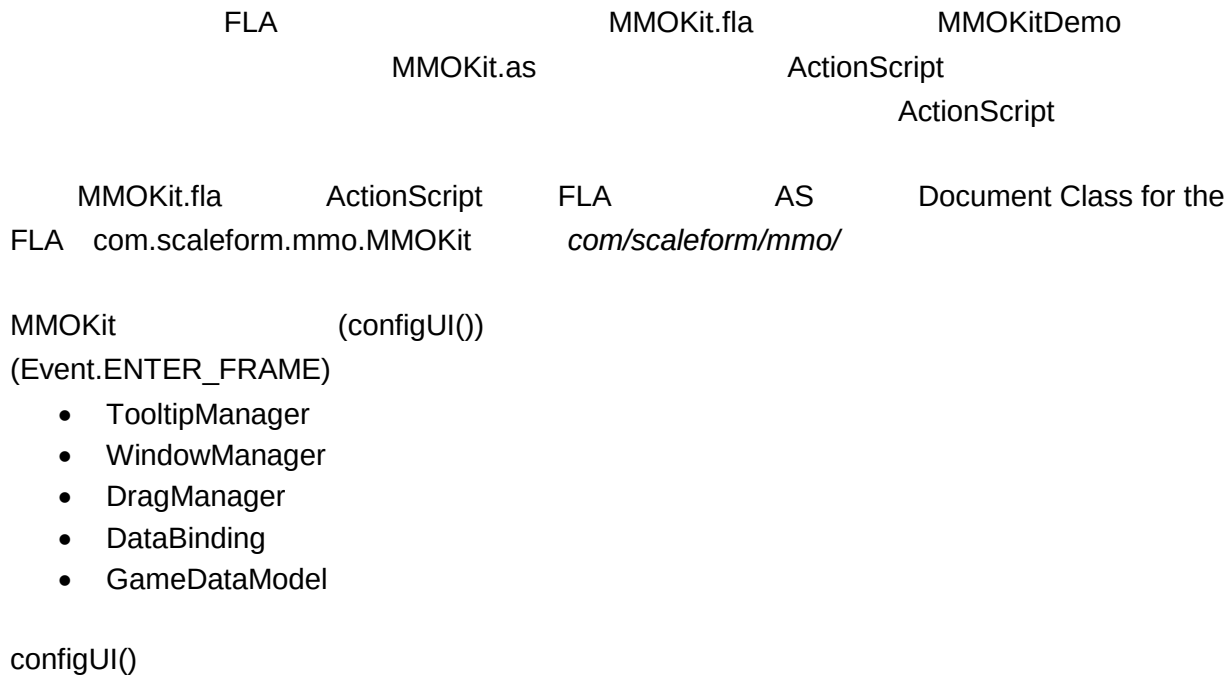
MMO Flash *Bin/Data/AS3/Kits/MMO/*

FlashDevelop  
MMOKit.as3proj ActionScript

UI FLA FLA  
FLA ActionScript  
*com/scaleform/mmo* ActionScript

Inventory Flash *widgets/InventoryView fla* Inventory  
AS *com/scaleform/mmo/inventory/* InventoryView.as Document Class

### 3.2.1 MMOKit fla



### 3.2.2

